

Benefits in tasks related to everyday life competences after a working memory training in older adults

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Objective: The impact of working memory (WM) training on everyday life functioning has rarely been examined, and it is not clear whether WM training gains are transferred to reasoning abilities. The aim of this study was to assess the efficacy of a verbal WM training in older adults, in terms of specific gains and transfer effects to everyday life and reasoning abilities.

Method: Thirty-six community dwelling older adults (from 65 to 75 years of age) were randomly assigned to a training or an active control group. The specific gains in a WM task similar to the one trained were assessed. Transfer effects to everyday life and reasoning abilities were also examined using (i) objective performance-based tasks (the Everyday Problem Test and the Timed Instrumental Activities of Daily Living Scale) and (ii) the Cattell test and Raven's Standard Progressive Matrices, respectively.

Results: Only the trained group showed specific benefits and transfer effects to one of the everyday abilities measures (the Everyday Problem Test) and in the two reasoning tasks.

Conclusion: These results suggest that WM training can positively impact cognitive functioning and, more importantly, older adults' abilities in everyday living.

Key words: working memory training; older adults; transfer effects; everyday abilities; intelligence; aging

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Introduction

Working memory (WM), which is the ability to retain and simultaneously process information for use in other cognitive tasks (Baddeley and Hitch, 1974), is involved in cognitive processes (e.g., reasoning) and in complex abilities also needed in everyday life. It undergoes a linear decline with aging (Borella *et al.*, 2008), which is exacerbated in age-related disorders, following a gradient from mild cognitive impairment to Alzheimer's disease (Gagnon and Belleville, 2011). A growing body of literature is thus focusing on intervention targeting this basic mechanism of cognition to sustain older adult functioning.

Studies on WM training in aging have generated positive results in terms of specific gains in the trained tasks. Training gains in untrained tasks (called transfer effects) have been found too, although their effect size was small (Karch and Verhaeghen, 2014). Some studies have shown large transfer effects, however, while others have found none (see Borella *et al.*, 2014 for a review).

Although WM training in older adults has theoretically been designed to prevent age-related cognitive decline and to sustain everyday autonomy, aspects of daily living have rarely been assessed. This shortcoming is rather surprising, especially considering that WM has a major influence on older adults' performance in everyday life. It could be due partly to the difficulty of finding instruments appropriate for use with healthy older adults, because most tools are devised for clinical populations rather than for normally aging individuals.

To our knowledge, only three WM training studies have assessed training-related gains in aspects partly related to everyday life in aging, examining attention in everyday contexts and related self-reported improvements (Richmond *et al.*, 2011), language comprehension (Carretti *et al.*, 2013), and the perceived ability to prevent cognitive failures (Brehmer *et al.*, 2012). Although these studies have reported positive results, the outcome measures considered are as follows: (i) susceptible to bias due to respondents' expectations

being self-reported (Brehmer *et al.*, 2012), and (ii) the tasks used are not sufficiently representative of real older adults' everyday activities (Richmond *et al.*, 2011; Carretti *et al.*, 2013). In Richmond *et al.* (2011), for instance, in one of the subtests in the Test of Everyday Attention (Ridgeway *et al.*, 1994), participants had to count up and down as they watched a series of pictures of 'elevator doors'—an activity not particularly relevant for most older adults. It is, thus, important to assess whether, in aging, WM training gains lead to benefits to deal with the 'real' complex problems of everyday life.

The efficacy of WM training also remains to be clarified as concerns the feasibility of improving reasoning abilities, which are also strongly involved in the demands of daily living. Some studies have found transfer effects to reasoning, while others have not (Karchach and Verhaeghen, 2014). In addition, the majority of such studies assessed transfer to reasoning abilities with only one task (often Raven's Standard Progressive Matrices), producing controversial results.

The present study aimed to examine the efficacy of a verbal WM training in a sample of community-dwelling older adults, seeking any transfer effects to everyday functioning and reasoning. For each of these two abilities, two indicators were used to clarify whether any changes affected the general ability (e.g., reasoning), or were only task-specific (e.g., von Bastian and Oberauer, 2013). To assess training gains in everyday functioning, rather than self-report measures, that is the Instrumental Activities of Daily Living Scale (IADLs; Lawton and Brody, 1969), performance-based measures of functioning that objectively assess the ability to solve everyday tasks were used: to this Everyday Problem Test (EPT; Willis and Marsiske, 1993), and the Timed Instrumental Activities of Daily Living Scale (TIADL; Owsley *et al.*, 2002). The EPT is a paper-and-pencil task that forse meglio "requires to solve" everyday problem (e.g., 'calculate the price of a product'), while the TIADL is a behavioral simulation task (e.g., 'find a number in a phone directory') that involves performing actions. The EPT was chosen because it has been shown to predict an individual's real functioning (Moore *et al.*, 2007), as measured by means of direct behavioral observation (Schmitter-Edgecombe *et al.*, 2011); and the TIADL because the time taken to complete everyday tasks is also thought to discriminate between normal and pathological aging, as between cases with mild cognitive impairment and healthy older adults (Wadley *et al.*, 2008).

The Cattell test and Raven's Standard Progressive Matrices were used to assess transfer effects to reasoning abilities.

The training administered was the verbal program developed by Borella *et al.* (2010). It was chosen because (i) it has generated promising results in terms of specific and transfer effects across different studies and populations (Borella *et al.*, 2014); (ii) the procedure is challenging and motivating for older adults; and (iii) it is appropriate for older adults because it involves only a few sessions.

Participants were randomly assigned to either a training group or an active control group. Consistently with previous results, we expected to obtain (i) specific effects on the so-called criterion task, that is a task very similar to the one trained, used to assess specific training gains (e.g., Borella *et al.*, 2010). Transfer effects are generally expected for tasks that involve the same processing resources as the task used in the training (e.g., Morrison and Chein, 2011). As WM has a well-documented role in everyday tasks (e.g., Allaire and Marsiske, 1999) and relationship with reasoning tasks (Borella *et al.*, 2006), training gains could be expected in both everyday tasks and reasoning skills.

Participants

Thirty-six healthy older adults from 65 to 75 years of age who were living in the community took part in the study. They were recruited on a voluntary basis through social clubs. They were all Italian mother-tongue.

Our inclusion criteria were a Mini-Mental State Examination score (Folstein *et al.*, 1975) ≥ 27 , indicating good cognitive functioning; and the absence (established from a semi-structured interview) of physical or psychological conditions capable of causing cognitive deficits according to Crook *et al.* (1986). At the pre-test, all participants performed above the cut-off for their age and education in the Categorization Working Memory Span task (CWMS; see Italian norms, De Beni *et al.*, 2008) and vocabulary test (Wechsler Adult Intelligence Scale—Revised, 1981; Italian norms by Orsini and Laicardi, 2003).

Participants were randomly assigned to the trained or active control groups. The groups did not differ in terms of age, years of formal education ($F_s < 1$), vocabulary test score, $F_{(1, 34)} = 2.27$, $p = 0.14$, or gender distribution (Table 1). The study was approved by the ethical board of the Padua University Faculty of Psychology.

Materials

Criterion task. Categorization Working Memory Span task (De Beni *et al.*, 2008). The task consisted of 10

sets of word lists, each of which included 20 lists of words that were divided into groups containing from two to six lists. Participants listened to a set of word lists audio-recorded at a rate of one word per second, and they had to tap with their hand on the table whenever an animal noun was heard (processing phase). The interval between word lists was 2 s. At the end of a set, participants recalled the last word on each list (maintenance phase), that is, they needed to remember from two to six words altogether, depending on the difficulty of the set. The total number of words recalled was used as the measure of WM performance (maximum 20).

Five of the sets were used as a pre-test version, the other five at the post-test stage. The two versions were counterbalanced across testing sessions.

Transfer effects. Everyday functioning measures. *Everyday problem solving (adapted by Borella and Cantarella et al. (in preparation) from the 42-item version of Willis and Marsiske (1993)).* This task measures the ability to solve challenging real-world cognitive problems embedded within instrumental domains. Each stimulus consists of two open-ended questions representing 'real-life' situations and covering instrumental activities (household management, health and use of medicines, shopping, managing finances, phone bills, transportation, and food preparation). Participants answered questions about how they would deal with these issues. Fourteen stimuli/situations most closely reflecting the Italian lifestyle were chosen from the original version; seven were used at pre-test, and seven at post-test, counterbalancing them among participants (maximum score 14).

Timed Instrumental Activities of Daily Living (adapted from Owsley et al. (2002) by Cantarella et al. (2016)). The task comprises five activities of daily living in the following IADL domains: communication, use of money, cooking, shopping, and use of medicine. Participants had to simulate the actions involved in each

domain (e.g., looking up a phone number), using the materials (real everyday objects) contained in an ad hoc kit, and time constraints were imposed (from 2 to 3 min, depending on the item). If the participant did not complete the task within the allotted time, testing on that particular activity was terminated.

The dependent variable was the time (in seconds) taken to complete the actions required for all the activities. It was adjusted by a time penalty calculated on the basis of the participant's accuracy in performing the tasks (Owsley *et al.*, 2002).

A parallel version was prepared, changing the stimuli to five activities of daily living. The two versions were distributed between the participants in a counterbalanced fashion.

Reasoning tasks. *Culture fair test (Cattell and Cattell, 1973).* This consists of two parallel forms, each containing four subtests (to be completed in 2.5 to 4 min, depending on the subtest) that involve (i) choosing from among six different options which ones correctly complete a series of figures; (ii) identifying figures or shapes that do not belong in a sequence; (iii) choosing items that correctly complete matrices of abstract figures; and (iv) assessing relationships between a series of items. The dependent variable is the number of correctly answered items across the four subtests (maximum score 50).

Parallel forms were administered in counterbalanced fashion across test sessions.

Raven's Standard Progressive Matrices (RSPM; Raven et al., 1977). Participants saw 3 × 3 matrices composed of abstract shapes, lines, and figures. They had to choose the item that best completed the matrix from six or eight possible options. The original 40-problem version was split to obtain two parallel versions (as in Shipstead *et al.*, 2012). The number of correct answers was the dependent variable (maximum score 20).

Procedure

The experimental procedure described here was in accordance with the Declaration of Helsinki (Sixth Revision, 2008). Participants in both groups attended five individual sessions: the first and fifth sessions, which lasted about 90 min each, were for the pre-test and post-test. During the other three sessions, the active control group was occupied in alternative 'activities' (Table 2), while the trained group was administered the WM training. For both groups, all activities were

Table 1 Demographic characteristics of the trained and control groups

	Trained group (N = 18; 50% Male)		Trained group (N = 18; 50% Male)	
	M	SD	M	SD
Age (years)	69.50	3.25	69.77	2.92
Education	9.94	2.50	10.50	3.09
Vocabulary	48.11	3.16	46.22	4.27

Note: M, mean; SD, standard deviation.

completed within a 2-week time frame, with a fixed 2-day break between sessions. The duration of the sessions and the amount of interaction with the experimenter were much the same for the two groups.

The three sessions of WM training (sessions 2–4) lasted about 30–40 min. Lists of words that were audio-recorded and organized as for the CWMS task were presented to the participants in the trained group, who were asked to recall target words, and tap on the table with their hand when they heard an animal noun. The maintenance demand of the CWMS task was manipulated by increasing the number of words that successful participants were asked to recall, and by presenting the lowest memory load to participants who were unsuccessful (session 2). The demands of the task also varied and, depending on the session, they could involve having to recall (i) the last or first word in each list and (ii) words that were preceded by a beep sound. The processing requirement (tapping on the table when an animal noun occurred) was also manipulated by varying the frequency of these animal words in the lists (session 3). Participants in the active control group were asked to fill in questionnaires (see Table 2 for details).

Statistical analyses

First analyses of variance (ANOVAs) were run with Group (trained, control) as the between-subjects factor on the participants' pre-test performance in all tasks. Results indicated that there were no baseline differences between the trained and control groups ($F_s < 1$ for all variables; Table 3). To assess training gains, 2 (Group: trained, control) $\times$ 2 (Session: pre-test and post-test) mixed-design ANOVAs were run on the measures of interest. Effect sizes for within-subject comparisons were computed using Cohen's d (Cohen, 1988); values were corrected using the Hedges and Olkin (1985) correction factor to avoid the small sample bias.

Results

Descriptive statistics are given in Table 3, and the results of the ANOVAs are summarized in Table 4.

Criterion task: CWMS. There was a main effect of Group, with trained participants correctly recalling more words than controls, and of Session, with a better

Table 2 Description of the training sessions by group

Session	Trained group	Control group
1. Pre-test 2. Training	CWMS task, Cattell test, EPT, Raven's Progressive Matrices, TIADL <i>WM training:</i> sets of three lists each containing from two to five words were used in three different phases. Participants had to recall target words (which changed from one phase to the next) and tap with their hand on the table whenever they heard an animal noun. The three phases of the WM task were presented one after the other: participants had to recall the last words on the lists in the 1st phase, the first words in the 2nd, and the last words again in the 3rd. In each phase, if a participant correctly recalled the target words for two of the three lists of the same length in a given set, the difficulty of the task was increased up until the lists contained five words each. Participants failing to complete the most difficult sets in a given phase started the next phase from the easiest level (i.e., the shortest word lists).	Autobiographic Memory questionnaire (De Beni <i>et al.</i> , 2008): participants were asked to remember common events related to their childhood, adulthood and recent events, and to rate their vividness.
3. Training	<i>WM training:</i> four sets of word lists of increasing difficulty were used (each containing from two to five lists). The sets of two-word lists could contain from 2 to 8 animal nouns altogether, the sets of three-word lists from 4 to 9, the sets of four-word lists from 6 to 11, and the sets of five-word lists from 8 to 17. For each set, participants had to tap with their hand on the table whenever they heard an animal noun and also remember each word followed by a beep sound, in serial order.	Memory Sensitivity (De Beni <i>et al.</i> , 2008): in this questionnaire, participants had to rate the frequency of their behavior dedicated to saving memories of life events.
4. Training	<i>WM training:</i> Four sets of two-word lists were used. Participants had to tap with their hand on the table whenever they heard an animal noun and they also had to recall: the last word in each list contained in the 1 st set presented; the first word in the 2 nd ; the last word in the 3 rd ; and the first word in the 4 th .	Psychological Well-being (De Beni <i>et al.</i> , 2008): participants were asked to rate personal satisfaction with life (past, present, and future), emotional competences (ability to understand one's own and other's emotions), and coping strategies regarding everyday problems.
5. Post-test	CWMS task, Cattell test, EPT, Raven's Progressive Matrices, TIADL	

Note: CWMS, Categorization Working Memory Span task; TIADL, Timed Instrumental Activities of Daily Life; EPT, Everyday Problem Solving.

performance at post-test than at pre-test. The significant Group $\times$ Session interaction indicated that trained participants performed better at post-test than at pre-test ($M_{diff}=2.56$, $p=0.001$; 95%CI [1.75; 3.36]). No significant differences across sessions emerged for the control group ($M_{diff}=0.11$, $p=0.78$; 95%CI [-0.91; 0.69]). At post-test level, the trained group performed significantly better than the control group ($M_{diff}=2.50$, $p=0.001$; 95%CI [1.70; 3.29]).

Transfer effects to everyday abilities. For the EPT, only the Group $\times$ Session interaction was significant: trained participants performed better at post-test than at pre-test ($M_{diff}=1.89$, $p=0.001$; 95%CI [0.83; 2.94]). No significant differences emerged across sessions for the control group ($M_{diff}=-0.66$, $p=0.20$; 95%CI [-0.71; 0.38]). At post-test level, the trained group performed significantly better than the control group ($M_{diff}=2.05$, $p=0.001$; 95%CI [1.00; 3.10]).

For the TIADL, in terms of the total time spent completing the activities, only the main effect of Session was significant: both groups improved in their post-test performance, taking less time to complete the task.

Transfer effects to reasoning tasks. For the Cattell test, only the main effect of Session and the Group $\times$ Session were significant. Performance was better at post-test than at pre-test ($M_{diff}=1.17$, $p=0.001$; 95%CI [1.66; 0.67]), and only the trained group consistently improved in performance from pre-test to post-test ($M_{diff}=2.06$, $p=0.001$; 95%CI [1.35; 2.75]). The control group's performance did not differ significantly between sessions ($M_{diff}=0.27$, $p=0.42$; 95%CI [-0.42; 0.97]). At post-test, the trained group performed significantly better than the control group ($M_{diff}=1.55$, $p=0.005$; 95%CI [0.50; 2.60]).

Using RSPM, the main effect of Session was significant (performance was better at post-test than at pre-test ($M_{diff}=1.08$, $p=0.002$; 95%CI [1.75; 0.42])), as well as the Group $\times$ Session interaction: only the trained group performed better at post-test than at pre-test ($M_{diff}=1.94$; $p=0.001$, 95%CI [1.00; 2.88]). No significant differences emerged for the control group ($M_{diff}=0.22$; $p=0.63$, 95%CI [-0.71; 1.15]), nor were there any significant differences between the two groups' post-test performance ($M_{diff}=1.12$, $p=0.05$, 95%CI [0.002; 2.33])).

Concerning effect size, the trained group revealed a large effect size (over 0.80) on the criterion task and in all transfer tasks except for the TIADL, for which the effect size was moderate (0.54). Only small effect sizes were found in the control group (Figure 1).

Discussion

The aim of this study was to ascertain whether WM training improves performance in everyday abilities and reasoning. Surprisingly, few studies have examined this issue; even though these abilities are important, they decline with aging, and they are strongly related to WM. Training gains seen on reasoning abilities have also been inconsistent across studies.

The particular novelty of the present study lies in that we selected tasks that include real-life stimuli to objectively examine transfer effects on healthy older adults' ability to solve everyday problems. Performance-based measures of accuracy in everyday functioning (EPT) and time of execution (TIADL) were used, and two indicators were selected for each of the constructs of interest (i.e., everyday abilities and reasoning).

In line with the literature and consistently with our expectations, we found a specific effect on the criterion

Table 3 Descriptive pre-test and post-test data by group

	Trained group		Control group		Trained group		Control group	
	Pre-test				Post-test			
	M	SD	M	SD	M	SD	M	SD
Criterion task								
CWMS words recalled correctly	8.72	1.27	8.66	1.61	11.28	1.12	8.78	1.21
Performance-based measures of functional abilities								
EPT	10.28	1.53	10.78	1.99	12.17	1.25	10.11	1.81
TIADL (time, ms)	436.51	84.94	424.01	107.96	391.57	77.67	400.23	88.26
Fluid intelligence tasks								
Cattell test	15.78	1.77	16.00	1.68	17.83	1.24	16.27	1.80
RSPM	9.89	2.52	10.44	2.68	11.83	1.72	10.67	1.71

Note: M, mean; SD, standard deviation CWMS, Categorization Working Memory Span task; EPT, Everyday Problem Solving; TIADL, Timed Instrumental Activities of Daily Living; RSPM, Raven's Standard Progressive Matrices.

Table 4 Mixed-design 2×2 ANOVA for the measures of interest, with group (trained versus control) as between-subjects factor and session (pretest versus post-test) as repeated measures

			$F_{(1, 34)}$	MSE	p^*	np^2
Criterion task: specific effect						
CWMS words recalled correctly	Between subjects	Group	14.23	2.06	0.001	0.29
	Within subjects	Session	22.61	1.41	0.001	0.40
		Group X Session	19.00	1.41	0.001	0.36
Transfer effects on performance-based measures of functional abilities						
EPT	Between subjects	Session	3.45	3.15	0.07	0.09
		Group	2.79	2.40	0.10	0.08
	Within subjects	Group X Session	12.20	2.40	0.001	0.26
TIADL	Between subjects	Group	0.00	11599.62	0.94	0.00
	Within subjects	Session	4.47	4748.48	0.04	0.11
		Group X Session	0.42	4748.48	0.51	0.01
Transfer effects on fluid intelligence tasks						
Cattell test	Between subjects	Group	1.85	4.32	0.18	0.05
	Within subjects	Session	22.96	1.06	0.001	0.40
		Group X Session	13.32	1.06	0.001	0.28
RSPM	Between subjects	Group	0.21	7.81	0.65	0.00
	Within subjects	Session	11.04	1.91	0.002	0.24
		Group X Session	6.97	1.91	0.01	0.17

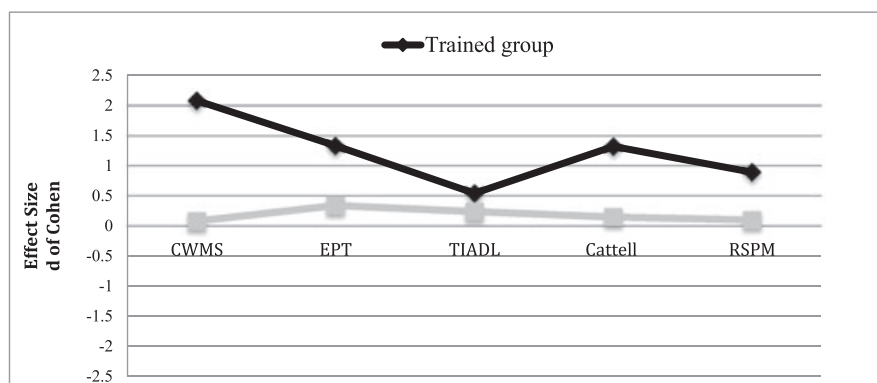
Note:

ANOVA, analysis of variance; F, Fisher–Snedecor statistic; MSE, mean standard error; $p = p$ -value; np^2 , partial eta-squared; CWMS, Categorization Working Memory Span task; EPT, Everyday Problem Solving; TIADL, Timed Instrumental Activities of Daily Living; RSPM, Raven's Standard Progressive Matrices.

*Significant main effects and interactions were analyzed using pairwise comparisons, with Bonferroni's adjustment for multiple comparisons. The α value was set at 0.05 for all statistical tests and at 0.006 for interactions.

task only in the trained group. Transfer effects to one of the everyday measures, the EPT, were also found. This latter result indicates that the WM training was effective in enhancing problem-solving performance in everyday life activities. No improvement was identified, however, in the other performance-based measure, the TIADL. This may be because the demands of the tasks were too easy for our participants (normally aging older adults).

In terms of accuracy, all participants performed well already at the pre-test, nearing a ceiling effect (92.12% of participants made no mistakes). The lack of any transfer effect due to a ceiling effect is in line with other studies that also failed to identify any training gains using another measure of everyday activities, the Test of Everyday Attention (Richmond *et al.*, 2011). On the other hand, this result is in contrast with the improvements



Note. CWMS=Categorization Working Memory Span task; EPT= Everyday Problem Test, TIADL: Timed Instrumental Activities of Daily Living; RSPM= Raven's Standard Progressive Matrices

Figure 1 Effect size d for pre-test and post-test comparisons of specific and transfer effects by group (trained participants versus controls). Note. CWMS, Categorization Working Memory Span task; EPT, Everyday Problem Test, TIADL, Timed Instrumental Activities of Daily Living; RSPM, Raven's Standard Progressive Matrices.

found in this measure in training studies focusing on processing speed (Ball *et al.*, 2002; Edwards *et al.*, 2002). These latter studies considered samples with initial processing speed difficulties, however, which is a different case from ours. In our study, although the effect did not reach statistical significance, the corresponding effect size was in the medium range for the trained group and small for the control group. In the light of these findings, future studies should make an effort to assess transfer effects on everyday competences by including measures such as the Observed Task of Daily Living, which are more appropriate than the TIADL in terms of the task's difficulty and demands—for healthy older adults at least. It would also be of interest to examine WM training transfer effects to other WM-related everyday abilities that might influence the capacity to live independently, such as spatial navigation skills (Meneghetti *et al.*, 2015).

As for transfer effects on reasoning tasks, consistently with previous studies using the same training procedure (e.g., Borella *et al.*, 2010, 2013; Carretti *et al.*, 2013), only the trained group showed an improvement in the Cattell test. The present results extend this benefit to the RSPM task too. This pattern of results confirms that WM is involved in abstract reasoning (Ackerman *et al.*, 2005; Borella *et al.*, 2006), and that they share much the same underlying processing resources—a prerequisite for transfer effects.

The present results thus support the previous evidence of the training procedure adopted here having an effect both on reasoning abilities (as partly demonstrated already in previous studies), and on everyday tasks.

A number of limitations of the present study need to be borne in mind. First, no follow-up assessment was conducted to see whether the training gains were maintained. The large effect sizes found and the fact that previous reports showed lasting effects of the same training procedure as was used here lead us to expect them, but there is no guarantee of any long-term effects. Future studies should thus examine maintenance effects of these gains. It is worth adding here that no follow-up analyses were conducted in more than half of the previously published WM training studies on older adults. Second, although our sample size was consistent with those used in other training studies (ranging from 30 to 57 older adults), these results would need to be replicated in larger samples.

It would also be of interest to assess training benefits in old-old individuals who are liable to a more accentuated decline in everyday competences (Schaie *et al.*, 2005). WM-related gains should also be examined in other populations with WM impairments and related

difficulties in everyday functioning (Malouin *et al.*, 2004), such as patients with brain injuries (Weicker *et al.*, 2015), stroke (Westerberg *et al.*, 2007), or schizophrenia (Wexler *et al.*, 2000). We have also examined training gains in a particular sample of older adults, that is, healthy young-old individuals characterized by a milder decline in cognitive plasticity than in late adulthood (Baltes and Smith, 2003).

The present study could thus be considered as a first, encouraging step forward in the assessment of everyday life benefits after a WM training. It suggests that WM training can be a promising approach for fostering older adults' autonomy in everyday living and quality of life, and thereby for promoting active aging.

Conflict of interest

None declared.

Key points

- Working memory (WM) training studies have generally not assessed transfer gains to daily life.
- The effect of WM training on reasoning abilities is not well established.
- WM training gains in measures of functional abilities and reasoning were examined.
- WM training-related gains were found in the Everyday Problem Test and reasoning tasks (Cattell test and Raven's Standard Progressive Matrices).
- WM training may be a valid way to sustain older adults' ability to live independently.

Author contributions

Erika Borella designed the study, supervised the data collection, carried out the statistical analyses, and wrote the paper. Barbara Carretti designed the study, supervised the data collection, assisted with analyzing data, and wrote the paper. Alessandra Cantarella supervised the data collection, carried out the statistical analyses and wrote the paper. Matthias Kliegel assisted with writing the article. Rossana De Beni assisted with writing the article.

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